

Microbiology Lessons





About the Course

An elective in Life Science, Microbiology is a challenging course for the Alveary senior. Some self-direction is required.

This topic is included in the following course(s): Science: Microbiology



Placement & Combining Tips

Students should take General Biology before Microbiology. Note that there is some sensitive content in this course, and it is recommended for students in Grade 12.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
12	Microbiology Lessons 5 times/week 45 min	A Field Guide to Bacteria A Planet of Viruses Growing a Revolution: Bringing Our Soil Back to Life I Contain Multitudes Micro Life Microbe Hunters The Immortal Life of Henrietta Lacks

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Microbiology				
Microbiology Lessons Nature Walks & Scouting: Grades 9-12	Microbiology Lessons	Microbiology Lessons	Microbiology Lessons Nature Notebook: Grades 9-12	Microbiology Lessons Microbiology Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Microbiology Lessons

- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.
- ☐ Gather supplies for the first microbiology labs initially. You will gather additional supplies once you decide on additional labs.

□ Teacher Note: Preread Chapter 15. Henrietta Lacks's children had to deal with a lot of abuse of every kind. Discuss with your student(s). The note for the student is in Week 1 of Term 2.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Microbiology Lessons



A Field Guide to Bacteria



A Planet of Viruses



Growing a Revolution: Bringing Our Soil Back to Life



I Contain Multitudes



Micro Life



Microbe Hunters



The Immortal Life of Henrietta Lacks



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

Microbiology Lessons

- ∞ [Nature Journal Connection](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Illustrated Guide to Home Biology Experiments](#)
- ∞ [Seven Wonders of the Microbe World](#)
- ∞ [Microbiology from A to Z](#)
- ∞ [Microbiology Current Events](#)

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or scan the QR
code for links.



Microbiology Lessons

How To Approach



Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

Microbiology Lessons

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Term 1

WEEK 1 45m Microbiology Lessons - Lesson 1

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 1 45m Microbiology Lessons - Lesson 2

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week, starting with the Appendices. Continue with the Introduction and read forward. Sometimes there will be practical things to do with the microscope and bacteria, but other times you will be reading to gain a greater understanding of bacteria from many different places. Sometime before next week, watch the author's video on bacteria. You may or may not agree with the author's perspective, but the point of the video is to see where some of the more recent thinking is concerning microbes.

→ VIEW

∞ Video Link: Talk by Betsey Dexter

WEEK 1 45m Microbiology Lessons - Lesson 3

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 1 45m Microbiology Lessons - Lesson 4

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

Gather some family and friends and watch the video "Seven Wonders of the Microbe World."

→ VIEW

∞ Video Link: Seven Wonders

WEEK 1 45m Microbiology Lessons - Lesson 5

Materials: Illustrated Guide to Home Biology Experiments

→ READ

Read p.3-51 at your own pace over the next few lab periods. If you already have a good working microscope of reasonable quality, you may be able to skim through the microscope chapter. If you are not that familiar with your microscope, you may benefit from reading it. You will move at your own pace through the microbiology labs in this book after completing the first 6 in order (Labs I-1 through II-3). Then, you are free to pick and choose based on your interests and/or the availability of needed supplies. The more you do, the more you will grow in your understanding of microbiology! There are several labs in the book that are general biology labs (10-12,18,20,21,23, and 24). You will skip over these and focus on the ones that require a microscope.

WEEK 2 45m Microbiology Lessons - Lesson 6

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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Term 1

WEEK 2 45m Microbiology Lessons - Lesson 7

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Come up with an interesting way to present or summarize the information in this book. It could be a map, chart, diagram, or something else.

WEEK 2 45m Microbiology Lessons - Lesson 8

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 2 45m Microbiology Lessons - Lesson 9

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 2 45m Microbiology Lessons - Lesson 10

Materials: Illustrated Guide to Home Biology Experiments

→ READ

Finish reading p.3-51.

WEEK 3 45m Microbiology Lessons - Lesson 11

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 3 45m Microbiology Lessons - Lesson 12

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 3 45m Microbiology Lessons - Lesson 13

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 3 45m Microbiology Lessons - Lesson 14

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 3 45m Microbiology Lessons - Lesson 15

Materials: Illustrated Guide to Home Biology Experiments

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Term 1

→ LAB DAY

WEEK 4 45m Microbiology Lessons - Lesson 16

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

nature walk - record observations

science free read

WEEK 4 45m Microbiology Lessons - Lesson 17

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 4 45m Microbiology Lessons - Lesson 18

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

Gather some family and friends and watch the video "How to Identify Microbes."

→ VIEW

∞ Video Link: How to Identify Microbes

WEEK 4 45m Microbiology Lessons - Lesson 19

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 4 45m Microbiology Lessons - Lesson 20

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 5 45m Microbiology Lessons - Lesson 21

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week

• PLAN WEEKLY

nature walk - record observations

science free read

WEEK 5 45m Microbiology Lessons - Lesson 22

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 5 45m Microbiology Lessons - Lesson 23

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

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Term 1

WEEK 5 ☐ 45m Microbiology Lessons - Lesson 24

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 5 ☐ 45m Microbiology Lessons - Lesson 25

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 6 ☐ 45m Microbiology Lessons - Lesson 26

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 6 ☐ 45m Microbiology Lessons - Lesson 27

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 6 ☐ 45m Microbiology Lessons - Lesson 28

☐ Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 6 ☐ 45m Microbiology Lessons - Lesson 29

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 6 ☐ 45m Microbiology Lessons - Lesson 30

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 31

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 32

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

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Term 1

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 33

☐ Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 34

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 35

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 36

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 37

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 38

☐ Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 39

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 40

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 41

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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Term 1

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 42

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 43

☐ Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 44

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 45

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 10 ☐ 45m Microbiology Lessons - Lesson 46

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 ☐ 45m Microbiology Lessons - Lesson 47

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 10 ☐ 45m Microbiology Lessons - Lesson 48

☐ Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 10 ☐ 45m Microbiology Lessons - Lesson 49

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 10 ☐ 45m Microbiology Lessons - Lesson 50

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

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Term 1

WEEK 11 45m Microbiology Lessons - Lesson 51

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 11 45m Microbiology Lessons - Lesson 52

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 11 45m Microbiology Lessons - Lesson 53

Materials: Microbe Hunters or A Planet of Viruses

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 11 45m Microbiology Lessons - Lesson 54

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 11 45m Microbiology Lessons - Lesson 55

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 12 45m Microbiology Lessons - Lesson 56

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 57

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 58

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 59

Exam Week

→ EXAMS

Answer question(s) related to course.

Microbiology Lessons

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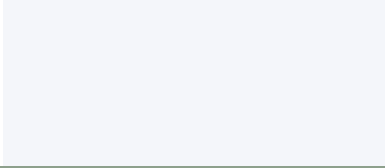
Term 1

WEEK 12 45m Microbiology Lessons - Lesson 60

Exam Week

→ EXAMS

Answer question(s) related to course.



Microbiology Lessons

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Term 2

WEEK 1	45m Microbiology Lessons - Lesson 61	Materials: I Contain Multitudes	→ READ, NARRATE, & DISCUSS Continue to read and narrate 8 pages each week. Challenge yourself to come up with a new way of narrating. Ask questions of the text and seek to find the answers. Discuss the ideas with others.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 1	45m Microbiology Lessons - Lesson 62	Materials: A Field Guide to Bacteria	→ READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 1	45m Microbiology Lessons - Lesson 63	Materials: Microbe Hunters or Micro Life	→ READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR skim-read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	Gather some family and friends and watch the video "Wonders of the Microbial World." → VIEW ∞ Video Link: Wonders of the Microbial World
WEEK 1	45m Microbiology Lessons - Lesson 64	Materials: The Immortal Life of Henrietta Lacks	NOTE: Sensitive content Chapter 15 contains disturbing accounts of child physical and sexual abuse. Discuss with your teacher before reading.	
WEEK 1	45m Microbiology Lessons - Lesson 65	Materials: Illustrated Guide to Home Biology Experiments	→ LAB DAY Continue lab work, writing up your report after completing each lab.	
WEEK 2	45m Microbiology Lessons - Lesson 66	Materials: I Contain Multitudes	→ READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY ☐ nature walk - record observations ☐ science free read
WEEK 2	45m Microbiology Lessons - Lesson 67	Materials: A Field Guide to Bacteria	→ READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 2	45m Microbiology Lessons - Lesson 68	Materials: Microbe Hunters or Micro Life	→ READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	

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Term 2

WEEK 2 45m Microbiology Lessons - Lesson 69 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 2 45m Microbiology Lessons - Lesson 70 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 3 45m Microbiology Lessons - Lesson 71 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY - nature walk - record observations - science free read
WEEK 3 45m Microbiology Lessons - Lesson 72 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 3 45m Microbiology Lessons - Lesson 73 Materials: Microbe Hunters or Micro Life → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	
WEEK 3 45m Microbiology Lessons - Lesson 74 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 3 45m Microbiology Lessons - Lesson 75 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 4 45m Microbiology Lessons - Lesson 76 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY - nature walk - record observations - science free read
WEEK 4 45m Microbiology Lessons - Lesson 77 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	

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Term 2

WEEK 4	45m Microbiology Lessons - Lesson 78 Materials: Microbe Hunters or Micro Life → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	
WEEK 4	45m Microbiology Lessons - Lesson 79 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 4	45m Microbiology Lessons - Lesson 80 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 5	45m Microbiology Lessons - Lesson 81 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY nature walk - record observations science free read
WEEK 5	45m Microbiology Lessons - Lesson 82 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages a week.	
WEEK 5	45m Microbiology Lessons - Lesson 83 Materials: Microbe Hunters or Micro Life → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	Gather some family and friends and watch something interesting from this YouTube channel. → VIEW ∞ Video Link: Journey to the Microcosmos
WEEK 5	45m Microbiology Lessons - Lesson 84 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 5	45m Microbiology Lessons - Lesson 85 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	

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Term 2

WEEK 6 45m Microbiology Lessons - Lesson 86

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 6 45m Microbiology Lessons - Lesson 87

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 6 45m Microbiology Lessons - Lesson 88

Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

WEEK 6 45m Microbiology Lessons - Lesson 89

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 6 45m Microbiology Lessons - Lesson 90

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 7 45m Microbiology Lessons - Lesson 91

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 7 45m Microbiology Lessons - Lesson 92

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 7 45m Microbiology Lessons - Lesson 93

Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

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Term 2

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 94

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 7 ☐ 45m Microbiology Lessons - Lesson 95

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 96

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 97

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 98

☐ Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 99

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 8 ☐ 45m Microbiology Lessons - Lesson 100

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 101

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 9 ☐ 45m Microbiology Lessons - Lesson 102

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

Microbiology Lessons

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Term 2

WEEK 9 45m Microbiology Lessons - Lesson 103 Materials: Microbe Hunters or Micro Life → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	Gather some family and friends and watch something interesting from this YouTube channel. → VIEW ∞ Video Link: Journey to the Microcosmos
WEEK 9 45m Microbiology Lessons - Lesson 104 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 9 45m Microbiology Lessons - Lesson 105 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 10 45m Microbiology Lessons - Lesson 106 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY - nature walk - record observations - science free read
WEEK 10 45m Microbiology Lessons - Lesson 107 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 10 45m Microbiology Lessons - Lesson 108 Materials: Microbe Hunters or Micro Life → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.	
WEEK 10 45m Microbiology Lessons - Lesson 109 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 10 45m Microbiology Lessons - Lesson 110 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	

Microbiology Lessons

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Term 2

WEEK 11 45m Microbiology Lessons - Lesson 111

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 11 45m Microbiology Lessons - Lesson 112

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 11 45m Microbiology Lessons - Lesson 113

Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

WEEK 11 45m Microbiology Lessons - Lesson 114

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 11 45m Microbiology Lessons - Lesson 115

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 12 45m Microbiology Lessons - Lesson 116

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 117

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 118

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 119

Exam Week

→ EXAMS

Answer question(s) related to course.

Microbiology Lessons

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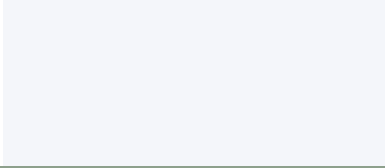
Term 2

WEEK 12 45m Microbiology Lessons - Lesson 120

Exam Week

→ EXAMS

Answer question(s) related to course.



Microbiology Lessons

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Term 3

WEEK 1 45m Microbiology Lessons - Lesson 121 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Continue to read and narrate 8 pages each week. Challenge yourself to come up with a new way of narrating. Ask questions of the text and seek to find the answers. Discuss the ideas with others.	• PLAN WEEKLY nature walk - record observations science free read
WEEK 1 45m Microbiology Lessons - Lesson 122 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 1 45m Microbiology Lessons - Lesson 123 Materials: Microbe Hunters OR Growing a Revolution → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.	Gather some family and friends and watch something interesting from this YouTube channel. → VIEW ∞ Video Link: Journey to the Microcosmos
WEEK 1 45m Microbiology Lessons - Lesson 124 Materials: The Immortal Life of Henrietta Lacks → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week. Discuss with your teacher.	
WEEK 1 45m Microbiology Lessons - Lesson 125 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY Continue lab work, writing up your report after completing each lab.	
WEEK 2 45m Microbiology Lessons - Lesson 126 Materials: I Contain Multitudes → READ, NARRATE, & DISCUSS Read and narrate 8 pages each week. Discuss with others.	• PLAN WEEKLY nature walk - record observations science free read
WEEK 2 45m Microbiology Lessons - Lesson 127 Materials: A Field Guide to Bacteria → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week.	
WEEK 2 45m Microbiology Lessons - Lesson 128 Materials: Microbe Hunters OR Growing a Revolution → READ, NARRATE, & DISCUSS Read and narrate 10 pages each week from Microbe Hunters OR about a	

Microbiology Lessons

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Term 3

chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 2 45m Microbiology Lessons - Lesson 129

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 2 45m Microbiology Lessons - Lesson 130

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 3 45m Microbiology Lessons - Lesson 131

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 3 45m Microbiology Lessons - Lesson 132

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 3 45m Microbiology Lessons - Lesson 133

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 3 45m Microbiology Lessons - Lesson 134

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 3 45m Microbiology Lessons - Lesson 135

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 4 45m Microbiology Lessons - Lesson 136

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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WEEK 4 45m Microbiology Lessons - Lesson 137

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 4 45m Microbiology Lessons - Lesson 138

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 4 45m Microbiology Lessons - Lesson 139

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 4 45m Microbiology Lessons - Lesson 140

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 5 45m Microbiology Lessons - Lesson 141

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 5 45m Microbiology Lessons - Lesson 142

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 5 45m Microbiology Lessons - Lesson 143

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

Gather some family and friends and watch something interesting from this YouTube channel.

→ VIEW

∞ **Video Link:** Journey to the Microcosmos

WEEK 5 45m Microbiology Lessons - Lesson 144

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

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WEEK 5 45m Microbiology Lessons - Lesson 145

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 6 45m Microbiology Lessons - Lesson 146

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 6 45m Microbiology Lessons - Lesson 147

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 6 45m Microbiology Lessons - Lesson 148

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 6 45m Microbiology Lessons - Lesson 149

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 6 45m Microbiology Lessons - Lesson 150

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 7 45m Microbiology Lessons - Lesson 151

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 7 45m Microbiology Lessons - Lesson 152

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 7 45m Microbiology Lessons - Lesson 153

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a

Microbiology Lessons

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chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 7 45m Microbiology Lessons - Lesson 154

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 7 45m Microbiology Lessons - Lesson 155

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 8 45m Microbiology Lessons - Lesson 156

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 8 45m Microbiology Lessons - Lesson 157

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 8 45m Microbiology Lessons - Lesson 158

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 8 45m Microbiology Lessons - Lesson 159

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 8 45m Microbiology Lessons - Lesson 160

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 9 45m Microbiology Lessons - Lesson 161

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

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WEEK 9 45m Microbiology Lessons - Lesson 162

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 9 45m Microbiology Lessons - Lesson 163

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

Gather some family and friends and watch something interesting from this YouTube channel.

→ VIEW

∞ Video Link: Journey to the Microcosmos

WEEK 9 45m Microbiology Lessons - Lesson 164

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 9 45m Microbiology Lessons - Lesson 165

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 10 45m Microbiology Lessons - Lesson 166

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

WEEK 10 45m Microbiology Lessons - Lesson 167

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 10 45m Microbiology Lessons - Lesson 168

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 10 45m Microbiology Lessons - Lesson 169

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

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WEEK 10 45m Microbiology Lessons - Lesson 170

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 11 45m Microbiology Lessons - Lesson 171

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 11 45m Microbiology Lessons - Lesson 172

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

WEEK 11 45m Microbiology Lessons - Lesson 173

Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

WEEK 11 45m Microbiology Lessons - Lesson 174

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

WEEK 11 45m Microbiology Lessons - Lesson 175

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 12 45m Microbiology Lessons - Lesson 176

Exam Week

→ EXAMS

Answer question(s) related to course.

• PLAN WEEKLY

- nature walk - record observations
- science free read

WEEK 12 45m Microbiology Lessons - Lesson 177

Exam Week

→ EXAMS

Answer question(s) related to course.

WEEK 12 45m Microbiology Lessons - Lesson 178

Exam Week

→ EXAMS

Answer question(s) related to course.

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Term 3

WEEK 12 45m Microbiology Lessons - Lesson 179 <i>Exam Week</i> → EXAMS Answer question(s) related to course.	
WEEK 12 45m Microbiology Lessons - Lesson 180 <i>Exam Week</i> → EXAMS Answer question(s) related to course.	

Microbiology Lessons

Examination

Term 1

- I Contain Multitudes: Describe some of the 20th-century discoveries in the field of microbes.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- A Planet of Viruses: Write about the history of the study of viruses. Include some comments on COVID-19 as part of the ongoing story after the book was written; OR Microbe Hunters: Write about the discoveries of Leeuwenhoek or Spallanzani.
- The Immortal Life of Henrietta Lacks: Tell the story of Henrietta Lacks (so far) in the book.
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.

Term 2

- I Contain Multitudes: Explain how microbes play a role in illness and in health.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- Micro Life: Write about a few of your favorite images. How do they help your understanding to see things up close? Include a few sketches to illustrate your points; OR Microbe Hunters: What role did Koch play in the history of microbes?
- The Immortal Life of Henrietta Lacks: What questions about bioethics have you had after reading the story of Henrietta Lacks to this point?
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.

Term 3

- I Contain Multitudes: Describe the future of microbe research.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- Growing a Revolution: Why is David Montgomery optimistic about agriculture? What are ways to leave the soil better off instead of degrading it?
- The Immortal Life of Henrietta Lacks: Write 5 response statements or 5 new questions about the story after finishing The Immortal Life of Henrietta Lacks.
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.